

SELF-PACED SWE SYSTEMS MASTERCLASS

GIT GOOD

The 365-Day SWE Playbook

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$ git commit -m "fix(career): might take 1 year to complete" --no-verify
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Written & Verified by me & Synthesized via AI

2026 · First Edition

About the Author and This Book

About the Author

I'm Srajan Rai, a software engineer writing backend systems in C#/.NET who got deeply tired of current job market and levels of interviews that demand everything from theory to actual hands-on experience.

Like most developers, I spent years writing code that was completely fine—right up until one day I got laid off and took some time off to react to this job famine. So after getting back from my vacation and preparing for 1 month 16 hrs a day, I got the job and wanted to create a self pace play book to help myself. I created this book to help myself and others like me. Every chapter in this 365-day masterclass exists because I refused to rely on buzzwords or memorized answers. I wanted to understand the absolute everything: how CPU pipelines execute instructions, why cache line false sharing degrades performance, how lock-free primitives eliminate contention, and how distributed state machines prevent split-brain anomalies under network partitions. And yes, I bought an iPad specifically to sketch every single memory layout and state machine in this book.

The standard was simple: if I couldn't explain the physical hardware mechanics, sketch the memory layout, trace the failure recovery path, and defend the trade-offs on a whiteboard under pressure, I didn't actually own the concept. This playbook is that roadmap.

About This Book

This is not a high-level cheat sheet, and it is not an academic textbook created by a human. It is an unapologetic, exhaustive **365-day, 12-volume systems curriculum** engineered to take me from any level to Staff-level systems design and architecture.

Every single module follows an unyielding **5-Layer Pedagogical Framework**:

- **1. Physical Memory & Hardware Reality:** What physical RAM, CPU registers, and 64-byte cache lines are doing under the hood during execution.
- **2. Abstract Theory & Mathematical Invariants:** Formal complexity bounds (O , Ω , Θ), proof-of-correctness invariants, and state transition models.
- **3. Production-Grade Implementation:** Complete, unabridged source code with zero-allocation optimizations, error handling, and memory safety.
- **4. Elite Algorithmic & Real-World Scenarios:** Core interview archetypes, non-trivial edge cases, and high-frequency failure scenarios.
- **5. Verification, Synthesis & Examination:** Self-assessment checklists and multi-question quizzes with full technical rationale to audit knowledge gaps.

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How to Use This PDF

This book is built by an AI, though annotated and content farmed by me.

The Whiteboard Pages

Every even-numbered page is a pure, solid white workspace. Use it to trace memory addresses, sketch failure state machines (e.g., worker failover, split-brain quorums), and write down your spoken whiteboard answers before checking the technical keys.

Color Legend

- Indigo, Key Idea:** The foundational hardware contract, invariant, or architectural pattern.
- Orange, Common Misconception:** Latent performance traps, race conditions, memory leaks, and flawed architectural assumptions.
- Pink, Interview Angle:** How this exact mechanism is tested during FAANG, HFT, and senior backend technical rounds.
- Green, Tip / Interview-Ready Answer:** Production takeaways, zero-allocation micro-optimizations, and crisp whiteboard phrasing.
- Violet, Quiz & Verification:** Deep-dive self-assessment quizzes with comprehensive multiple-choice, writing prompts, and answer keys.

How to Take Notes

The whiteboards are not for rewriting what is printed. They are for synthesizing mechanics.

Restate the Physical Invariant

If a section explains CPU cache line false sharing or Raft log matching, sketch the multi-core or cluster topology yourself. Trace where the contention or quorum partition occurs.

Mark Failure Modes & Gaps

Whenever an invariant or edge case feels non-obvious (e.g., a node failing mid-transaction with an uncommitted write), place a prominent **?** on the whiteboard page. Re-verify the step-by-step state transitions.

Cross-Reference Across Volumes

When studying database storage engines in Volume X, cross-reference back to B-Tree cache line alignment in Volume II and asynchronous I/O in Volume VI. Senior systems engineering connects low-level primitives to high-level distributed systems.

Rehearse Spoken Explanations

Answer the Pink Interview boxes out loud without looking at notes. Write your exact spoken explanation on the whiteboard page. If you cannot articulate the trade-offs cleanly out loud, you do not fully own the concept yet.